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Syllabus
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DATA STRUCTURES
ALGORITHMS
COMPUTER NETWORKS
DATABASE SYSTEMS

Major Course: Computer Animation

Level	Semester	Course Code	Course Name	Credits	Teaching Hours	Exam Duration	Max Marks
5.5	VI	**112222**	**Computer Animation**	2	30	2 Hrs.	30

Unit System	Contents	Workload Allotted	Weightage of Marks Allotted	Incorporation of Pedagogies
Course Objectives:	1. To understand the basics of computer graphics and animation. 2. To learn animation principles and multimedia concepts. 3. To create 2D and basic 3D animations using animation tools. 4. To understand animation applications in media and entertainment.			
Course Outcomes:	Students will be able to - 1. understand the fundamentals of computer animation. 2. use animation tools and techniques. 3. create simple 2D and basic 3D animations. 4. understand real-world applications of animation.			
Unit I	Introduction to Computer Graphics and Animation: History and Evolution of Animation, Types of Animation**,** Traditional Animation, 2D Animation, 3D Animation, Stop Motion Animation, Principles of Animation, Multimedia Concepts, Applications of Computer Animation	8 Hrs.	8 Marks	Use any pedagogical technique suitable for the topic
Unit II	Introduction to Animation Software, Interface and Tools of Animation Software, Creating Shapes and Objects, Working with Layers and Frames, Timeline and Keyframes, Basic Drawing and Coloring Techniques, Image Editing Basics	7 Hrs.	7 Marks	
Unit III	Creating 2D Animation, Motion Tweening and Shape Tweening, Text Animation, Audio and Video Integration, Introduction to 3D Animation, Basic 3D Objects and Transformations, Lighting and Camera Basics, Rendering Concepts	8 Hrs.	8 Marks	
Unit IV	Character Animation Basics, Storyboarding Techniques, Visual Effects Basics, Animation in Gaming and Films, Web Animation Basics, Applications of Animation in Education, Advertising, and Entertainment, Recent Trends in Computer Animation	7 Hrs.	7 Marks	

References:	**Course Material/Learning Resources** **Text books:** 1. Computer Graphics and Animation – Pradeep K. Sinha 2. Multimedia and Animation – R. P. Singh 3. Computer Graphics – Donald Hearn and M. Pauline Baker **Reference Books:** 1. The Animator’s Survival Kit – Richard Williams 2. 3D Animation Essentials – Andy Beane 3. Fundamentals of Computer Graphics – Peter Shirley 4. Introduction to Computer Graphics – James D. Foley 5. Multimedia Systems – John F. Koegel Buford **Weblink to Equivalent MOOC on SWAYAM if relevant:** 1. https://onlinecourses.swayam2.ac.in/e-learning/preview/cec21_cs18 2. https://onlinecourses.nptel.ac.in/e-learning/preview/noc24_cs82 3. https://onlinecourses.swayam2.ac.in/e-learning/preview/ntr25_ed64 **Any pertinent media (recorded lectures, YouTube, etc.)if relevant:** 1. https://www.youtube.com/watch?v=NZbrdCAsYqU 2. https://www.youtube.com/watch?v=wibh3x3h03I 3. https://www.youtube.com/watch?v=N-HFQeZvaP0&t=275s 4. https://www.youtube.com/watch?v=uDqjIdI4bF4 5. https://www.youtube.com/watch?v=EFmxPMdBqmU 6. https://www.youtube.com/watch?v=B5RYDDgAMP4&list=PLV9FdvTG6uyRtoAl-any4kOaRZYOmHozS

Assessment Rubric: Internal for Theory Course (Major)

SANT GADGE BABA AMRAVATI UNIVERSITY, AMRAVATI THEORY EXAMINATION B.Sc. III, SEMESTER – VI (NEP)			
Theory	**Course Code:** 112222	**Course Title:Computer Animation**	**Max Marks: 20**
Sr. No.	**Assessment Criteria**		**Marks**
1	Any Two of the Activity (weekly or monthly assignment/ Unit Test/Class Test) **(CAT-1)**		**10**
2	Any One of the Activity includes Open Book Test, MCQ Tests, / Theory related Class activity/Task etc.) etc. **(CAT-2)**		**5**
3	Participation in any one Activity includes Quiz, Seminar, Group Discussion, Industry visit, Case Study, Mini project, innovative work, etc. based on Theory Course. Involvement in subject related activities such as Conference, Workshop, Coding Competitions etc. **(CAT-3)**		**5**